Institute of Geographical Information Systems
School of Civil & Environmental Engineering

National University of Sciences and Technology, Islamabad, Pakistan

Open Ended Lab	
Semester: 4th BE-2024	**GIE-203 Digital Mapping and Image Processing**
Time Allowed: 2 weeks	**Marks/ Score: 10**

Topic: Use of Spectral Indices for Flood Mapping (CLO2, PLO4)

Objectives:
To enable students to search and acquire freely available satellite imagery, select suitable pre- and post-flood scenes, and apply appropriate spectral techniques to map and assess flood extent and its impact on different land uses.

Relevance to Real Life (Practical Application):
This open-ended lab (OEL) activity will help students explore geospatial data portals that provide freely available satellite imagery from different remote sensing systems. Students will learn to select appropriate temporal datasets for flood analysis and generate flood inundation maps using spectral indices and image processing techniques. Such information can support disaster response, impact assessment, and planning for flood-prone regions in Pakistan.

Problem Statement:
Pakistan has recently experienced significant flooding events that have severely affected various regions, particularly in parts of **Punjab and Khyber Pakhtunkhwa (KPK)**. As a remote sensing analyst, you are tasked with conducting a flood mapping exercise for any **flood-affected area of your choice** within these provinces. Using freely available satellite imagery, you are required to download **pre- and post-flood datasets** for your selected area and generate a **flood inundation map**. The analysis should focus on identifying the spatial extent of floodwaters, quantifying the affected area, and highlighting key patterns or impacts observed in the imagery.

Task:
As an image analyst, you must provide accurate and useful information to the client regarding the spatial distribution of floodwaters and their impact. The client may be looking for the following response:

Required Response:
a. Selection and acquisition of freely available pre- and post-flood remote sensing imagery for the chosen area.
b. Application of suitable indices (e.g., NDWI, MNDWI) to delineate flood-inundated areas.
d. Comparison of pre- and post-flood conditions to map and quantify flood extent.
e. Estimation of the total inundated area (in Acres/Hectares).
f. Additional spatial analysis (e.g., proximity to settlements, affected land cover, or temporal change patterns).

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Deliverables:
Share the output in the form of maps and a written report covering important aspects of the operations you performed and the analysis of the outputs. The report must cover the following aspects:

a. Brief description of the assigned task and the selected/used data.
b. Identification of the appropriate processing technique.
c. Purpose and significance of the applied operations.
d. Evaluation and analysis of the final results.

Submission: 31st October 2025